

Coding & Solution

Date	30 June 2026
Team ID	xxxxxx
Project Name	OptiCrop
Maximum Marks	5 Marks

Coding & Solution Summary

Field	Details
Repository Link / URL	https://github.com/tanuboddhisivasaisrisachinreddy- archfp/OptiCrop-Smart-Agricultural-Production-OptimizationEngine
Programming Language(s)	Python, JavaScript, HTML, CSS
Framework(s) Used	Flask, Bootstrap, Machine Learning
Key Features Implemented	1. Crop recommendation using Machine Learning 2. User-friendly web interface 3. Soil and environmental parameter analysis 4. Input validation and error handling 5. Database connectivity for user management
Pending / Incomplete Features	1. Real-time weather API integration 2. Multi-language support 3. Mobile application support
Setup / Run Instructions	1. Install Python and required libraries 2. Open project folder in VS Code / Anaconda 3. Run the Flask application 4. Open localhost URL in browser

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop project follows clean coding practices with modular design and reusable components. Proper validation, documentation, and structured implementation improve maintainability and scalability of the application.